

Section 51.

MathScienc

#4. $\text{irr}(i, \mathbb{Q}) = x^2 + 1 (= (x+i)(x-i) \text{ in } \overline{\mathbb{Q}})$

$\text{irr}(\sqrt[3]{2}, \mathbb{Q}) = x^3 - 2 (= \prod_{k=0,1,2} (x - \sqrt[3]{2} \cdot \exp(2\pi i \cdot \frac{k}{3}) \text{ in } \overline{\mathbb{Q}})$

Take $\alpha \in \mathbb{Q}$ satisfying

$\alpha \neq \frac{\sqrt[3]{2}}{2i} \cdot (-1 + \exp(2\pi i \cdot \frac{k}{3})) \quad (k=1,2), \frac{\sqrt[3]{2}}{2}$

$= \frac{\sqrt[3]{2}}{2i} (-\frac{3}{2} + \frac{\sqrt{3}i}{2}) \text{ or } \frac{\sqrt[3]{2}}{2i} (-\frac{3}{2} - \frac{\sqrt{3}i}{2})$

$\frac{\sqrt{2}, \sqrt{3}}{\sqrt{2} + \sqrt{3}}$

Let $\alpha = 1$, and $\alpha = \sqrt[3]{2} + i$. Denote $\sqrt[3]{2} = \zeta$. ($\zeta^3 = 2$, clearly)

$\alpha^2 = \zeta^2 - 1 + 2i\zeta = (-1, 0, 1, 0, 2, 0)$

$\alpha^3 = 2 - i + 3\zeta^2\zeta - 3\zeta = (2, -3, 0, -1, 0, 3) \quad \alpha^4 = (1, \zeta, \zeta^2, i, i\zeta, i\zeta^2)$

$\alpha^4 = 2\zeta - i\zeta + 6i - 3\zeta^2 + 2i + (-3\zeta^2 - 3\zeta i) = (1, 2, -6, 8, -4, 0)$

$\alpha^5 = (-8, 4, 0, 1, 2, -6) + (-12, 1, 2, 0, 8, -4) = (-20, 5, 2, 1, 10, -10)$

(Since $[\mathbb{Q}(\sqrt[3]{2}, i) : \mathbb{Q}] = 6$, α^n ($n \geq 6$) are meaningless).

The set $\{\alpha^n \mid 0 \leq n \leq 5\}$ is linearly independent, therefore, we can find two polynomials in α as

$i = (a_0, 0, 0, 1, 0, 0) = \sum_{n=0}^5 a_n \alpha^n$

where $a_n, b_n \in F$. n^3

$\sqrt[3]{2} = (0, 1, 0, 0, 0, 0) = \sum_{n=0}^5 b_n \alpha^n$

Take $A = \begin{pmatrix} 1 & [\alpha]_F & [\alpha^2]_F & \dots & [\alpha^5]_F \end{pmatrix} \in M_{6 \times 6}(F)$

$A \begin{pmatrix} a_0 \\ \vdots \\ a_5 \end{pmatrix} = i = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \Leftrightarrow \begin{pmatrix} a_0 \\ \vdots \\ a_5 \end{pmatrix} = A^{-1} \cdot \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$ (Specifically, $(a_0, \dots, a_5) = \frac{1}{22} \cdot (9, 100, 18, 40, -9, 12)$)

and

$A \begin{pmatrix} b_0 \\ \vdots \\ b_5 \end{pmatrix} = \zeta = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \Leftrightarrow \begin{pmatrix} b_0 \\ \vdots \\ b_5 \end{pmatrix} = A^{-1} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$

Finally, calculating A^{-1} , we obtain the result.

$\hookrightarrow$

Alternatively, take $\alpha = \sqrt[3]{2} \cdot i$. Then, $\sqrt[3]{2} = \alpha^4 \cdot \frac{1}{2}$ and $i = -2\alpha^3$, thus

$F(\alpha) = F(\sqrt[3]{2}, i)$

$\underbrace{\hspace{2cm}}$

$\underbrace{\hspace{2cm}}$

#11. Algebraic extension of perfect is perfect.

K
 $\downarrow$
 E
 $\downarrow$
 F

Proof. Let F be a perfect field and E/F be an algebraic extension.

Let K/E be a finite extension of E .

Suppose that K is not separable over E . Then,

$$\{[K:E]\} < [K:E].$$

This implies that for some $\alpha \in K$, $\text{irr}(\alpha, E)$ has common multiplicity larger than 1.

Meanwhile, since K/E finite and E/F algebraic, $\alpha \in K$ is algebraic over F .

But, since F is a perfect field, $\text{irr}(\alpha, F)$ is separable over F .

But!! $\text{irr}(\alpha, E)$ divides $\text{irr}(\alpha, F)$, it is contradiction.

#12. Let $F \subseteq E \subseteq K$, E/F algebraic, K/E algebraic.

Let $\alpha \in K$ be given. Since α is algebraic over E and E/F algebraic, α is algebraic over F . Therefore, $\text{irr}(\alpha, F)$ exists.

By the assumption, $\text{irr}(\alpha, E)$ have common multiplicity as 1. Hence

$$\text{irr}(\alpha, E) = \prod a_n x^n \quad \text{where } a_n \in E.$$

Since $\text{irr}(\alpha, E)$ is contained in $F(a_0, \dots, a_{\deg \alpha})$ with multiplicity 1,

α is separable over $F(a_0, \dots, a_{\deg \alpha})$, which implies that

α is separable over F . ($F(a_0, \dots, a_{\deg \alpha})$ is separable, by the assumption)

#13. Let $F_s = \{\alpha \in E \mid \alpha \text{ is separable over } F\}$

Clearly $F \subseteq F_s$. Let $\alpha, \beta \in F_s$ be given.

Since $F(\alpha + \beta) \subseteq F(\alpha, \beta)$ and $F(\alpha, \beta)$ is separable over F , $F(\alpha + \beta)$ is separable.

Similarly $F(\alpha\beta)$ and $F(\alpha/\beta)$ ($\beta \neq 0$) too.

#14.

a). For any $\alpha \in E$, $\sigma_p^n(\alpha) = \alpha^{p^n} = \alpha$, because the finite field of order p^n is set of zeros of $x^{p^n} - x$.
If order of σ_p is smaller than n , then it is contradiction to the above fact.

b). Since E is finite field, $E/\mathbb{Z}_p$ separable and simple extension.

And E is splitting field of $\mathbb{Z}_p$, because it is generated by all roots of $x^{p^n} - 1$.

Now, $|G(E/\mathbb{Z}_p)| = [E:\mathbb{Z}_p] = n$ with

$\sigma_p \in G(E/\mathbb{Z}_p)$ has order of n ,

$$G(E/\mathbb{Z}_p) = \langle \sigma_p \rangle.$$

$$\sqrt[3]{2}, \sqrt[3]{2} \cdot \frac{-1 + i\sqrt{3}}{2}, \sqrt[3]{2} \cdot \frac{-1 - i\sqrt{3}}{2}$$

[Section 53,

#4. Clearly, $[K:\mathbb{Q}(\sqrt[3]{2}, i)] = 2$. Thus $|G(\mathbb{Q}(\sqrt[3]{2}, i))| = 2$.

#7. Similarly, $[K:\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3})] = 2$, gives that $|G(\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3}))| = 2$.

(Since $[\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3}):\mathbb{Q}] = [\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3}):\mathbb{Q}(\sqrt[3]{2})] \cdot [\mathbb{Q}(\sqrt[3]{2}):\mathbb{Q}] = 4$,

$$[K:\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3})] = 2.)$$

#11. Find $\text{Gal}(\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3})/\mathbb{Q})$. Denote $\zeta_k = \sqrt[3]{2} \cdot \exp(2\pi i \frac{k}{3})$ $k=1, 2$

a). $\sigma_0 = \text{identity}$. σ_1 $\sqrt[3]{2}$ maps to ζ_1 and $i\sqrt{3}$ maps to $i\sqrt{3}$ 3
 σ_2 " to ζ_2 and " 2
 σ_3 " to $\sqrt[3]{2}$ and $i\sqrt{3}$ maps to $-i\sqrt{3}$ 2
 σ_4 " to ζ_1 and " 3
 σ_5 " to ζ_2 and " 2

b). By the Cauchy's Theorem, group of order 6 is $\mathbb{Z}_6$ if it is Abelian, and S_3 if it is non-Abelian.

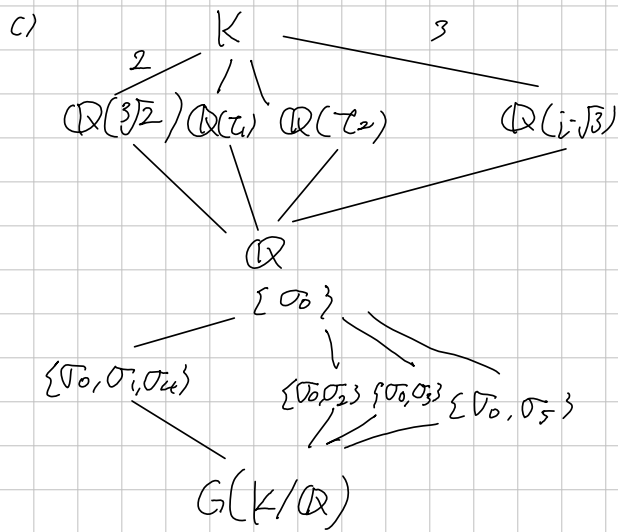
Enough to show $G(K/\mathbb{Q})$ is not Abelian. Observe that

$$(\sigma_1 \circ \sigma_3)(\sqrt[3]{2}) = \sigma_1(\sqrt[3]{2}) = \zeta_1 \text{ but}$$

$$(\sigma_3 \circ \sigma_1)(\sqrt[3]{2}) = \sigma_3(\zeta_1) = \sqrt[3]{2} \cdot (-1 - i\sqrt{3})/2 \neq \zeta_1. \text{ Thus } G(K/\mathbb{Q}) \text{ is not Abelian.}$$

$$\sigma_1^2(\sqrt[3]{2}) = \sigma_1(\zeta_1) = \zeta_1.$$

$$\sigma_5^2(\sqrt[3]{2}) = \sigma_5(\zeta_2) =$$



#6. Since $G(K/E) \trianglelefteq G(K/F)$,

$$G(E/F) \cong G(K/F) / G(K/E)$$
 and $G(K/F)$ is abelian $\Rightarrow$ the $G(E/F)$ is abelian.
 $G(K/E)$ is abelian, because it is subgroup of abelian.

Thm. Let K be a finite normal extension of F , and $F \leq E \leq K$.
 Then K is a finite normal extension of E and
 $\text{Aut}(K/E) \leq \text{Aut}(K/F)$.

$$\begin{array}{c} K \\ \vdots \\ E \\ \vdots \\ F \end{array}$$

Moreover, two Automorphisms σ and τ in $\text{Aut}(K/F)$ are satisfied $\sigma|_E = \tau|_E$
 if and only if
 $\sigma \text{Aut}(K/E) = \tau \text{Aut}(K/E)$

Proof. Above two assertions are trivial. Let $\sigma, \tau \in \text{Aut}(K/F)$ be given.

$$\begin{aligned} \sigma \text{Aut}(K/E) = \tau \text{Aut}(K/E) &\Leftrightarrow \tau^{-1}\sigma = \mu \text{ for some } \mu \in \text{Aut}(K/E) \\ &\Leftrightarrow \sigma|_E = \tau|_E \end{aligned}$$

Def. If K is a finite Normal Extension of F , then $\text{Aut}(K/F)$ is called
 Galois Group of K over F .

Galois Theory.

Let K/F be a finite normal extension.

$$\lambda: \{E \mid F \leq E \leq K\} \rightarrow \text{Gal}(K/F): E \mapsto \text{Gal}(K/E)$$

$$\begin{aligned} \text{For any } F \leq E \leq K, \quad [K:E] &= |\text{Gal}(K/E)|, \\ [E:F] &= |\text{Gal}(K/F) : \text{Gal}(K/E)| \end{aligned}$$

E is Normal of $F \Leftrightarrow G(K/E)$ is Normal subgroup of $G(K/F)$.

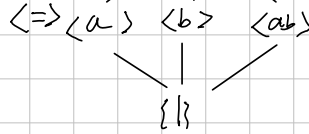
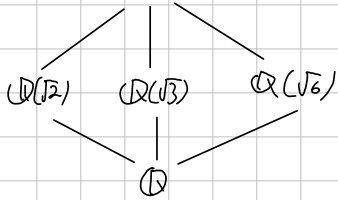
If E is Normal of F , then

$$G(E/F) \cong G(K/F) / G(K/E)$$

$$\mathbb{Q}(\sqrt{2}, \sqrt[3]{2})$$

$$\sqrt{2} \cdot \sqrt[3]{2}$$

$$\mathbb{Q}(\sqrt{2}, \sqrt{3}) = \mathbb{Q}(\sqrt{2+\sqrt{3}})$$



$$\text{irr}(\sqrt{2}, \mathbb{Q})$$

$$\text{irr}(\sqrt[3]{2}, \mathbb{Q})$$

$$\alpha = \sqrt{2} + \sqrt{3}$$

$$\alpha^2 = 5 + 2\sqrt{6}$$

$$\alpha^3 = 5\sqrt{2} + 4\sqrt{3} + 5\sqrt{3} + 6\sqrt{2} = 11\sqrt{2} + 9\sqrt{3}$$

$$(\alpha^3 - 11\alpha) \cdot -\frac{1}{2} = \sqrt{3}$$

$$(\alpha^3 - 9\alpha) \cdot \frac{1}{2} = \sqrt{2}$$

$$(a_1 + \dots + a_m)^n$$

$$= (a_1 + (a_2 + \dots + a_m))^n$$

$$= \sum \binom{n}{k_1} a_1^{n-k_1} \cdot (\dots)^{k_1}$$